

Exercise 15

Chapter 2, Page 76

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Introduction to Electrodynamics

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Let $\mathcal{S}_r$ denote the sphere of radius r centered at the origin and let Q_r be the total charge enclosed by $\mathcal{S}_r$. Since the charge distribution is spherically symmetric, it follows that the electric field is radially oriented and that it depends only on r (If you are, by any chance, feeling uneasy accepting this argument, consider the result from problem 2.7. or 2.11. The thick spherical shell is a superposition of thin layers. Every such layer generates a radially oriented field. Then, by the principle of superposition, the total field has to be radially oriented and cannot depend on θ or ϕ , since none of the fields from the layers did.)

Step 2

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If $r < a$, $Q_r = 0$ and if $r > b$, Q_r equals the total charge Q , which equals Q_b . For $a \leq r \leq b$ we have

$$\begin{aligned}
 Q_r &= \int_{\mathcal{S}_r} \rho \, d\tau \\
 &= 4\pi \int_a^r \rho(s) \, s^2 \, ds \\
 &= 4\pi \int_a^r \frac{k}{s^2} \, s^2 \, ds \\
 &= 4\pi k(r - a) .
 \end{aligned}$$

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By the Gauss's law, we have

$$\begin{aligned}
 \oint_{S_r} \mathbf{E} \cdot d\mathbf{a} &= \oint_{S_r} E da \\
 &= 4\pi r^2 E(r) \\
 &= \frac{1}{\epsilon_0} Q_r \implies \\
 E(r) &= \frac{1}{4\pi\epsilon_0} \frac{Q_r}{r^2} \\
 &= \begin{cases} 0 & \text{if } r < a \\ \frac{k}{\epsilon_0} \frac{r-a}{r^2} & \text{if } a \leq r \leq b \\ \frac{k}{\epsilon_0} \frac{b-a}{r^2} & \text{if } r > b \end{cases}
 \end{aligned}$$

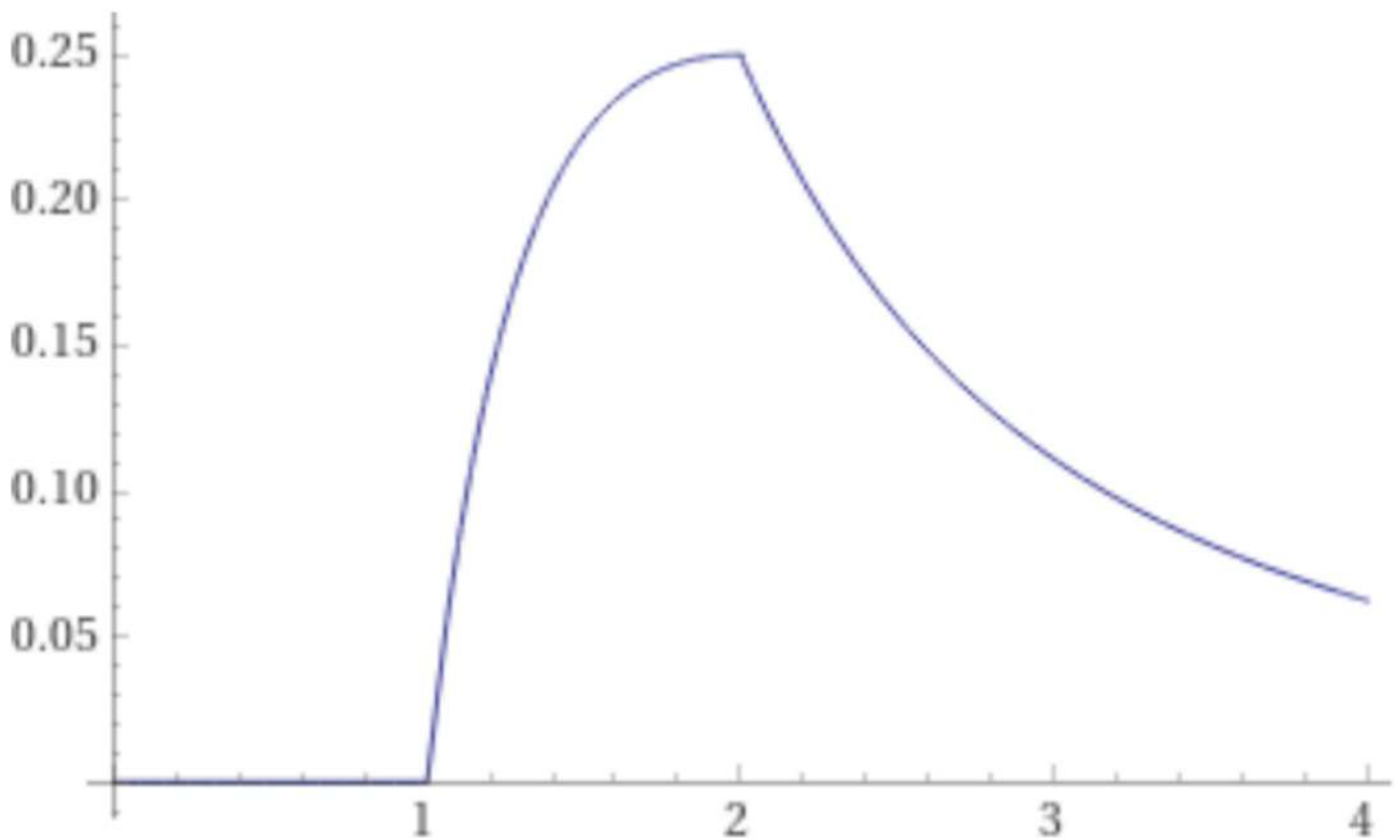
Step 4

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In the next box we have plotted the dependance of the field E on the distance from the origin r , for $k = \epsilon_0 = a = 1$ and $b = 2$.

Step 5

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$$E(r) = \begin{cases} 0 & \text{if } r < a \\ \frac{k}{\epsilon_0} \frac{r-a}{r^2} & \text{if } a \leq r \leq b \\ \frac{k}{\epsilon_0} \frac{b-a}{r^2} & \text{if } r > b \end{cases}$$

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